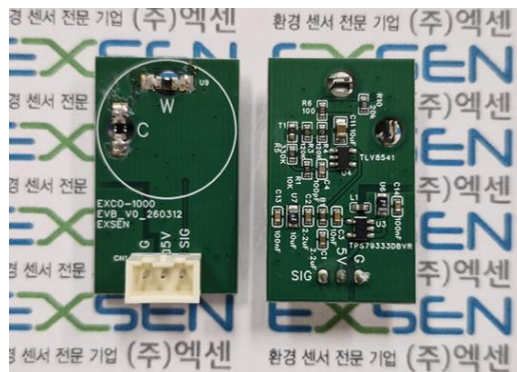


USER MANUAL

MODEL NAME	CO SENSOR EXCO-1000-2E EVB
PART NUMBER	EVB_EXCO_1000-2E_V0
CUSTOMER NAME	-
CUSTOMER PART NUMBER	-
DATE	2026.03.26
REMARK	R00
SOFTWARE VERSION	V0
SOFTWARE CHECKSUM	NONE



EXSEN Inc.

	Written	Reviewed	Approved
Signature			
Name			
Date			

Customer

	Written	Reviewed	Approved
Signature			
Name			
Date			

Contents

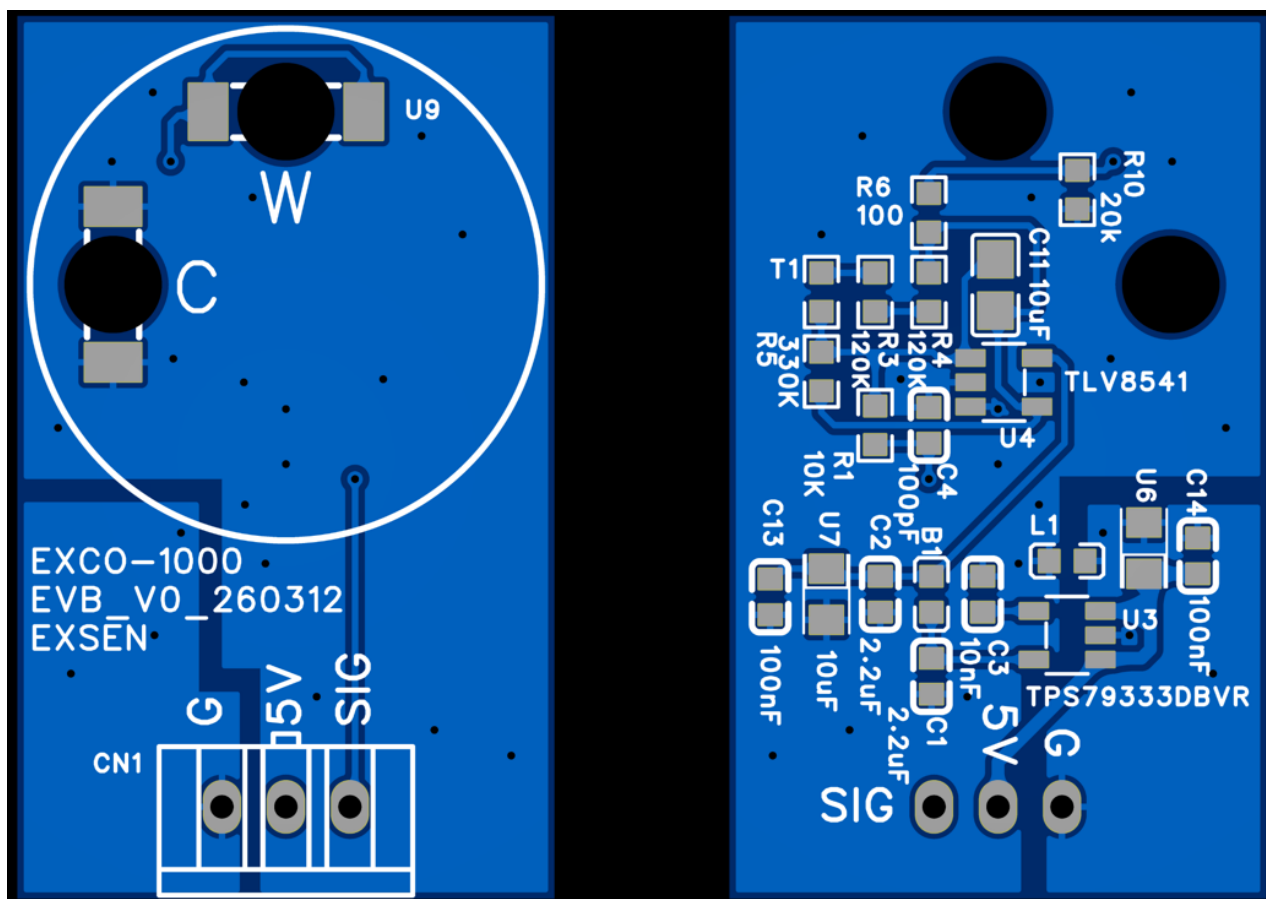
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2. DATA SHEET

MODEL NAME	CO SENSOR EXCO-1000-2E EVB
PART NUMBER	EVB_EXCO_1000-2E_V0
DIMENSION	21 x 34
DETECTION RANGE	0~500 ppm
COMMUNICATION	On Board TIA and LDO
APPLICATION	Carbon monoxide concentration measuring General Purpose

(1) Dimension

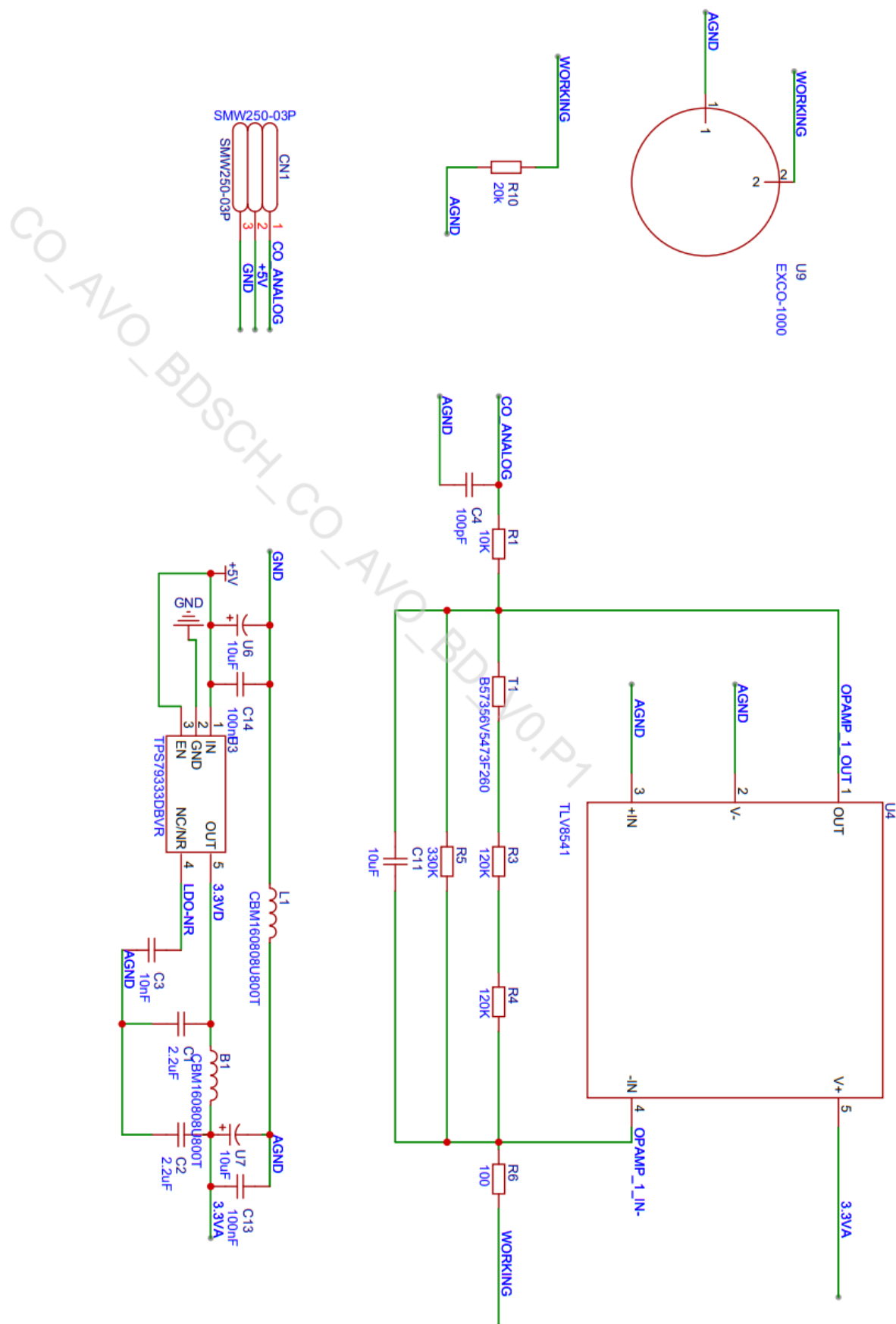
- 21 x 34 (W x H, mm)



General Tolerance (mm)

Linear	±0.3
Radius	±0.5

(2) Schematic Diagram



(3) BOM

No.	Quantity	Comment	Designator	Footprint	Value	Note
1	2	CBM160808U800T	B1,L1	L0603	80ohm	-
2	2	2.2uF	C1,C2	C0603	2.2uF	-
3	1	10nF	C3	C0603	10nF	-
4	1	100pF	C4	C0603	100nF	-
5	1	10uF	C11	C0805	10uF	-
6	2	100nF	C13,C14	C0603	100nF	-
7	1	SMW250-03P	CN1	SMW250-03P	3P	YEONHO
8	1	10K, 1%	R1	R0603	10K	F
9	2	120K, 1%	R3,R4	R0603	120K	F
10	1	330K, 1%	R5	R0603	330K	F
11	1	100, 1%	R6	R0603	100	F
12	1	20k	R10	R0603	20k	J or F
13	1	B57356V5473F260	T1	R0603	47k, 4050K	TDK
14	1	TPS79333DBVR	U3	SOT-23-5	3.3V	TI
15	1	TLV8541	U4	SOT-23-5		TI
16	2	10uF	U6, U7	CAP-SMD_L2.0-W1.3-FD	10uF	Tantal
17	1	EXCO-1000-2E	U9	CO_PCD13_5		EXSEN CO SENSOR
18	2	S9091-46R	U9	S9091-46R, EXCO-1000-2E footprint	-	HARWIN

(4) Sensor & electrical performance specification ($T_a = 25^\circ\text{C}$, 50%RH)

Parameters		Condition	Symbol	Min	Typ	Max	Unit
Power	Input	Bias is not required	V_{IN}	4.5	5	5.5	V
	Current Consumption	-	P_A	-	10	-	mA
Electrical	Interface connections	-	O_C	Voltage Output			V
	Sampling interval	-	T_{SPL}	-	Depends on receiver's system		Hz
	Connector	-	CNT	3pin, SMW-250-3P			-
Transimpedance Gain(Rf)				153.5			$k\Omega$
Temperature Compensation				Integrated NTC thermistor (T1)			

Note: The actual gain may vary by +3~5% depending on component tolerances and ambient temperature

(5) Pin Configuration

Pin No	Name	Description
1	SIG	CO Signal (Analog V)
2	5V	5V in
3	G	Ground
CNT	SMW-250-03P	

(6) CO concentration conversion equation

$$Sig(V) = I(sensor) \times Rf$$

$$R3 = 120k\Omega, R4 = 120k\Omega, R5 = 330k\Omega, T1 = 47k\Omega(@25^{\circ}C)$$

$$Rf = \frac{(R3 + R4 + T1) \times R5}{(R3 + R4 + T1) + R5}$$

$$\rightarrow Rf \approx 153.5k\Omega$$

$$I(sensor) = CO\ Conc(ppm) \times Sensitivity(\frac{nA}{ppm})^{note\ 1}$$

$$CO\ Conc(ppm) = \frac{Sig(V)}{sensitivity(\frac{nA}{ppm}) \times Rf(\Omega)}$$

Example)

$$if\ Sig(V) = 0.768\ V, sensitivity = 10\ nA/ppm$$

$$CO\ Conc(ppm) = \frac{0.768\ (V)}{10\ (\frac{nA}{ppm}) \times 153.5(k\Omega)}$$

$$V(V) = I\ (A) \times R(\Omega)$$

$$CO\ Conc(ppm) = \frac{0.768\ (V)}{1.535\ (\frac{mV}{ppm})}$$

$$\rightarrow CO\ Conc(ppm) \approx 500\ (ppm)$$

Note 1: The sensor sensitivity is indicated via a QR code and text printed on the side of the CO sensor body.

3. SAMPLE CODE

(1) Download site: <https://github.com/EXSEN/EXCO-1000-2E>

Notice

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